



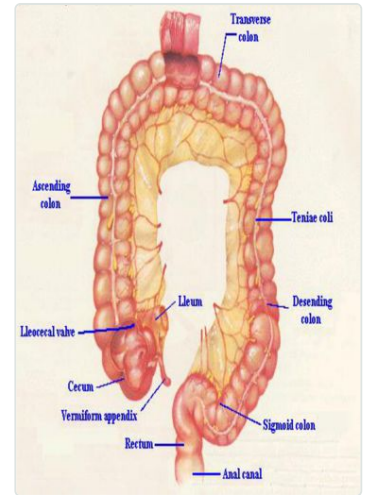
BEFORE YOU READ – 3 BLIND ID CHALLENGES

1. List five histological features that identify the large intestine.
2. Which layer forms the taeniae coli, and which two segments lack them?
3. Describe the epithelium in each zone of the anal canal from top to bottom — where is the dentate line?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 · Recognising the large intestine (ileocecal valve → anus)

- **Parts:** cecum (+ appendix) → colon (1.5 m: ascending, transverse, descending, sigmoid) → rectum → anal canal.
- **Five ID points:** numerous goblet cells; deep crypts, no villi, no folds; shorter, wider lumen than the small intestine; numerous lymphatic nodules; taeniae coli. The crypts of Lieberkühn are **long and straight**.
- Mucosa has no folds except in the rectal portion (vertical folds = columns of Morgagni). Epithelium: **simple columnar**, invaginated to form simple tubular glands (crypts).



Large intestine: cecum, appendix, colon, rectum, anal canal.

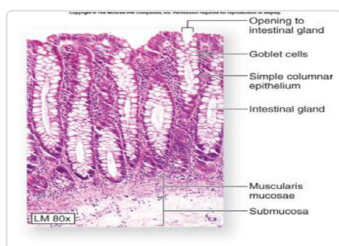
LOOK-ALIKE GRID – SMALL VS LARGE INTESTINE (SGL 2 VS SGL 3)

	SMALL INTESTINE	LARGE INTESTINE
Mucosal surface	villi + plicae circulares (jejunum)	no villi, no folds (rectal columns aside)
Crypts	simple tubular glands below the base of the villi	deep, long, straight crypts
Goblet cells	increase toward the ileum	numerous throughout
Lymphoid tissue	Peyer's patches in the ileum	numerous lymphatic nodules in the lamina propria (more than in the small intestine)
Outer longitudinal muscle	outer longitudinal layer	gathered into 3 thick bands = taeniae coli

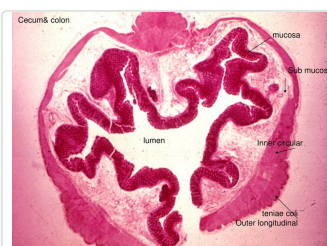
2 · Cecum, colon and rectum — cells and layers

EPITHELIAL CELL	WHERE	NOTE
Absorptive (columnar)	surface + crypts	more numerous
Goblet	surface + crypts	mucin granules → lubricant
Stem	base of crypts	turn over in 3-4 days ; precursors of the other cells
Endocrine	mainly the lower half of crypts	granules at the base of the nucleus
COVERING		SEGMENTS
Serosa — intraperitoneal		cecum, transverse colon, sigmoid colon
Adventitia & serosa — retroperitoneal		ascending and descending colon
Adventitia		rectum (histology like the rest of the large intestine but no taeniae coli)

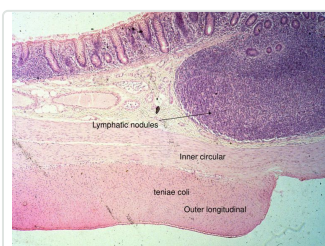
MAG	WHAT TO LOOK FOR
×4	No villi, no folds: parallel, deep crypts; outer muscle gathered into taeniae coli ; lymphatic nodules in the lamina propria.
×10	Crypts long and straight down to the muscularis mucosae; inner circular + outer longitudinal layers.
×40	Goblet cells packed with mucin among columnar absorptive cells; stem cells at the crypt base.



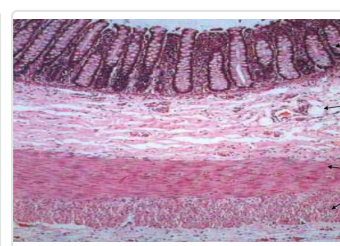
Colonic mucosa: intestinal glands, goblet cells, simple columnar epithelium, muscularis mucosae.



Cecum & colon: mucosa, submucosa, lumen, inner circular, taeniae coli.



Lymphatic nodules; taeniae coli.

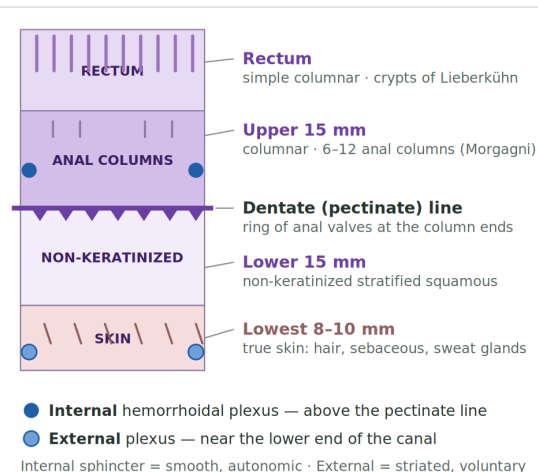


Rectum: mucosa, submucosa, muscularis externa, adventitia.

3 · Anal canal (~4 cm) — zones, plexuses, sphincters

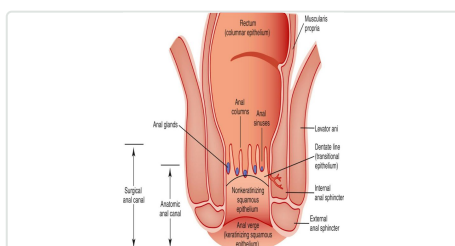
ZONE	LINING
Upper 15 mm	columnar ; 6–12 longitudinal anal columns (Morgagni); their lower ends joined by anal valves
Dentate line	anal valves form a complete ring = dentate (pectinate) line
Lower 15 mm	non-keratinized stratified squamous
Lowest 8–10 mm	true skin : hair follicles, sebaceous and sweat glands

- **Submucosa:** prominent venous plexus. **Internal** hemorrhoidal plexus lies **above** the pectinate line; **external** plexus lies near the lower end of the canal.
- **Muscle:** circular + longitudinal layers continue from the rectum; the circular layer thickens into the **internal anal sphincter** (**smooth** muscle, **autonomic**); outside it the **external anal sphincter** is **striated**, **voluntary**.

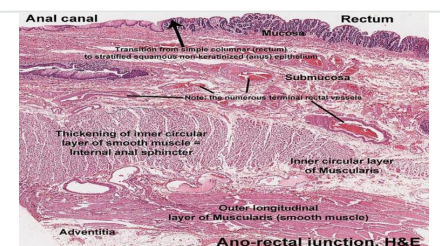


Anchor diagram: the anal canal, rectum at the top.

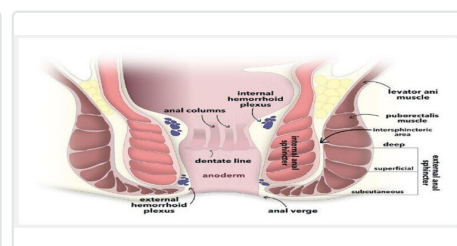
Slide wording: the lecture calls the vertical folds "rectal columns of Morgagni" at the recto-anal junction and later "anal columns (Morgagni)" in the upper anal canal — same columns. It also says the dentate line divides the canal into upper two-thirds and lower one-third, which does not match its own 15 / 15 / 8–10 mm zoning; learn the epithelium by zone.



Anal canal: columns, dentate line (transitional epithelium), sphincters.



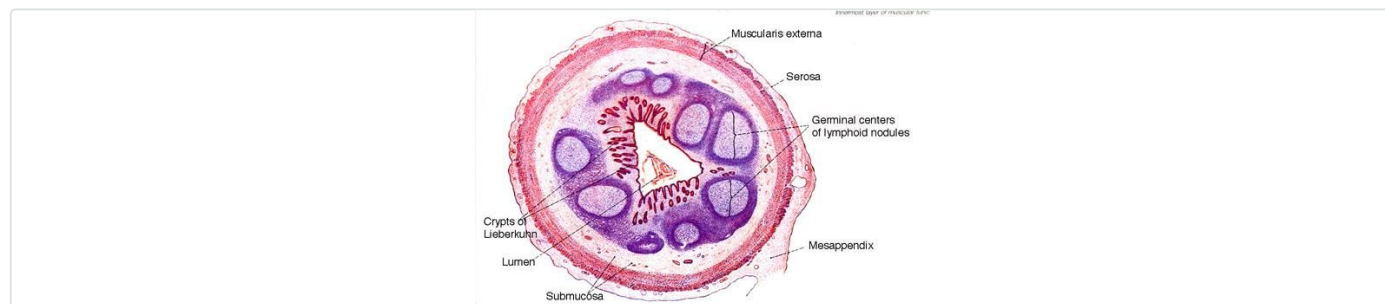
Ano-rectal junction (H&E): columnar → non-keratinized squamous.



Sphincters and hemorrhoidal plexuses.

4 • Appendix

	APPENDIX	COLON
Shape	small blind-ending evagination of the cecum ; 5–10 cm long, 0.8 cm across	—
Lumen	small, narrow, irregular	wide
Wall	abundant lymphatic nodules	nodules present
Glands	fewer, shorter	long, straight crypts
Outer muscle	no taeniae coli ; outer longitudinal + inner circular	outer longitudinal gathered into 3 taeniae



Appendix: crypts, lumen, submucosa, germinal centres of lymphoid nodules, muscularis externa, serosa, mesoappendix.

Pearls — what must survive if everything else is forgotten

- **Large intestine ID:** no villi, no folds, deep long straight crypts, numerous goblet cells, numerous lymphatic nodules, **taeniae coli** (3 thick bands of the outer longitudinal layer).
- **Rectum** = like the colon but **no taeniae coli**, covered by adventitia. **Appendix** = narrow irregular lumen, abundant nodules, fewer shorter glands, no taeniae.
- Covering: cecum / transverse / sigmoid = **serosa**; ascending / descending = **adventitia & serosa**; rectum = **adventitia**.
- **Anal canal (~4 cm):** upper 15 mm columnar (6–12 anal columns) → **dentate line** → lower 15 mm non-keratinized squamous → lowest 8–10 mm skin.
- **Internal** hemorrhoidal plexus above the pectinate line, **external** near the lower end. **Internal sphincter** = smooth, autonomic; **external** = striated, voluntary.

Test yourself — transfer-task MCQs

Unseen fields, not slide recall. Commit to an answer before you uncover the key.

1. A biopsy shows deep, straight crypts full of goblet cells, no villi, and an outer muscle layer gathered into three thick longitudinal bands. Which segment?

- A. Ileum
- B. Duodenum
- C. Rectum
- D. Appendix
- E. Colon

2. A blind-ending tube 8 cm long has a narrow irregular lumen, abundant lymphatic nodules in its wall, fewer and shorter glands than the colon, and no taeniae coli. What is it?

- A. Sigmoid colon
- B. Appendix
- C. Rectum
- D. Cecum
- E. Ileum

3. During proctoscopy a surgeon notes 6–12 longitudinal mucosal folds lined by columnar epithelium in the upper anal canal. What are they?

- A. Plicae circulares
- B. Rugae
- C. Anal columns (of Morgagni)
- D. Taeniae coli
- E. Anal valves

4. A dilated venous plexus is found in the submucosa above the pectinate line. Which plexus is it?

- A. External hemorrhoidal plexus
- B. Internal hemorrhoidal plexus
- C. Meissner's plexus
- D. Auerbach's plexus
- E. Myenteric plexus

5. After a nerve injury a patient loses voluntary control over defecation. Which structure is affected and how is it built?

- A. Internal anal sphincter — smooth muscle
- B. External anal sphincter — striated muscle
- C. Taeniae coli — smooth muscle
- D. Muscularis mucosae — striated muscle
- E. Internal anal sphincter — striated muscle

ANSWERS — COVER UNTIL YOU'VE COMMITTED

- 1. **E** — No villi + deep straight crypts + goblet cells + taeniae coli = colon (rectum and appendix lack taeniae).
- 2. **B** — These are the lecture's features of the appendix (5–10 cm, narrow irregular lumen, abundant nodules, no taeniae).
- 3. **C** — The upper 15 mm of the anal canal is columnar epithelium with 6–12 longitudinal anal columns (Morgagni).
- 4. **B** — The internal hemorrhoidal plexus lies above the pectinate line; the external plexus lies near the lower end of the canal.
- 5. **B** — The external anal sphincter is striated and voluntary; the internal is thickened smooth circular muscle under autonomic control.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

Histology GIT Block · MEDucation by Mattin Majid · Source: GIT Histology SGL 3, Dr Hero K. Mustafa
Study alongside the lecture, not instead of it.